

Community Ecology and Ecosystems BI2034 – 2026

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Realfagbygget
(or message through
Canvas)



Overview



Course introduction



Details and logistics for field trip



Introduction to community & ecosystems ecology
Lecture based on some recap on Norwegian nature

Ch 1 Mittelbach 2012 Community ecology
(History of the field)

And some extra slides on “how to do ecological studies Marchetti et al 2023 Ch 1



Bente J Graae



Sigurd Einum



Thor Harald Ringsby



Hanna Lee



Martijn Vandegehuchte



Snædur Adalsteinsson



Olena Mesleshko



Jane M. Hagen



Who are you?

- Name
- Where you come from
- One “fun fact” or hobby



Literature

- Begon and Townsend 2021 Ecology
- Richard Bardgett 2005. Biology of the Soil
- Articles and chapters from some other books

Begon and Townsend is available in library, and at Akademika. Other literature will be uploaded on Canvas



Schedule - Canvas

- Sessions
 - Mondays 8.15-10 F4
 - Tuesdays 14.15-16 R54
 - Wednesdays 12.15-14.00 V20 and 14.15-16 HB314
 - and Fridays 8.15-10.00 R3 and 12.15-14.00 HB314
- 15 double lectures
- 5 exercises
- Projects – fieldwork, labwork, data analysis, report writing
 - In groups of 5 – 7 students
 - 2 seminars presenting to class for discussions and inputs and seminars for presenting field course practicalities, questions to curriculum etc
- Report submission November 13th

Schedule BI2034 - Community Ecology and ecosystems 2026

Week	Day	Date	Time	Type	Teacher	Topic
34	Tue	18.aug	14.15-16.00	Lecture 1	BJG	Intro to course - Introduction to community ecology
	Wed	19.aug	08.15-10.00	Lecture 2	BJG	Patterns in community composition in space and time
	Fri	21.aug	08.15-10.00	Seminar	SE/BJG	Intro to field courses
	Fri	21.aug	12.15-14.00	Lecture 3	THR	Measuring biodiversity: Abundance and diversity of species
	35	Mon	24.aug	08.15-10.00	Lecture 4	SE
	Tue	25.aug	08.15-10.00	Lecture 5	SE	Abiotic factors in aquatic environments
Field course Sletvik						
	Wed	26/8	09.00-12.00	Bus to Sletvik from Realfagbygget Meet in front of main entrance to Realfagbygget		
	Wed	26/8	12.00-12.30	Lunch		
	Wed	26/8	13.00-20.00	Project prep	BJG, OM, PMP	Intro to project, outdoor demonstration, group forming, Preparing for Thursday's fieldwork
	Wed	26/8	17.00-18.00	Dinner	BJG, OM, PMP	
	Thur	27/8	09.00-17.00	Project	All	Fieldwork. Data gathering
	Thur	27/8	18.00-22.00	Project	All	Lab work. Sample processing and preparation
	Fri	28/8	0900-22.00	Project	All	Lab work. Sample processing and preparation
	Sat	29/8	9.00-13.00	Project start		Defining research questions, planning work ahead.
	Sat	29/8	14.00	Bus departs to Realfagbygget - arrival c. 16.00		
Field course Snåsa						
	Wed	26.aug	0815	Meet in DU2-150		
	Fri	28.aug	1200	Return to Realfagbygget		
36	All week	All week	All week	Project	PMP, JH, OM	Open lab work - participation to be scheduled
	Mon	31.aug	8.15-10.00	Lecture 6	HL	Soils and plant-soil interaction
	Tues	01.sep	14.15-16.00	Lecture 7	THR	Community models and struct. of ecol. communities
	Wed	02.sep	14.15-16.00	Lecture 8	PMP	Multivariate descriptions of communities
	Fri	04.sep	08.15-10.00	R/Statistics	SE	Crash course in R (R Studio)
37	Mon	07.sep	8.15-10.00	Lecture 9	HL	Flow of energy
	Tue	08.sep	14.15-16.00	Lecture 10	MV	Below and aboveground
	Wed	09.sep	12.15-14.00	R/Statistics	SE/PMP	Statistical methods: Group-specific excercises
	Fri	11.sep	08.15-10.00	R/Statistics	SE/PMP	Statistical methods: Group-specific excercises
	Fri	11.sep	12.15-14.00	Lecture 11	BJG	Plant traits-based community and ecosystems ecology
38	Tue	15.sep	14.15-16.00	Lecture 12	BJG	Patterns in biodiversity and their conservation
	Wed	16.sep	12.15-14.00	R/Statistics	SE/PMP	Project work: analyze own data
	Fri	18.sep	12.15-14.00	R/Statistics	SE/PMP	Project work: analyze own data
39	Mon	21.sep	8.15-10.00	Seminar	BJG	How to write a project report - project start

	Tues	22.sep	14.15-16.00	Project work		Do your project work	
	Wed	23.sep	12.15-16.00	Project work		Do your project work	
	Fri	25.sep	14.15-16.00	Project work		Do your project work	
40	Mon	28.sep	8.15-10.00	Seminar	All	Presentation of project questions and methods by each group	
	Tue	29.sep	14.15-16.00	Lecture 13	HL	Flow of matter	Ch 21 B&T
41	Mon	05. oct	8.15-10.00	Lecture 14	BJG	Ecology in a changing world	Ch 22 + Own lit
	Wed	07. oct	14.15-16.00	Lecture 15	BJG	Geoengineering and other big scale ecological changes	Ch 22 + Own lit
39-42	All week			Project		Project work - rooms available in exercise/seminar hours	
42	Wed	14.Oct	12.15-16.00	Seminar	All	Presentation of project results and tentative discussion	
44	Fri	23.Oct		Project work		Submission of draft report to group supervisors	
45	Fri	30.Oct		Project		Draft report feedback from group supervisors	
46	Fri	13.nov		Project		Hand in final project report	
51	Tue	15. dec	14.15-16.00	Recap	BJG, SE, HL	Questions to curriculum	
52	Tue	22.dec	09.00-13.00	Exam		Written exam	

Red text = Compulsory activities

Teachers

BJG	Bente Jessen Graae
SE	Sigurd Einum
OM	Olena Meleshko
HL	Hanna Lee
THR	Thor Harald Ringsby
MV	Martijn Vandegehuchte
PMP	Pauline Mesterton Pitter
SA	Snæthor Adalsteinsson
JH	Jane Hagen

Rooms:

Monday	8.15-10.00	F4
Tuesday	14.15-16.00	R54
Wednesday	12.15-14.00	VE20
Wednesday	14.15-16.00	GL314
Friday	8.15-10.00	R3
Friday	12.15-14.00	GL314

Literature: B&T: Begon and Townsend 2021. Own literature will be supplied on Canvas as PDFs

Literature for each lecture is also given in the schedule

Schedule BI2034 - Community Ecology and ecosystems 2020

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34	Tue	18.aug	14.15-16.00	Lecture 1	BJG	Intro to course - Introduction to community ecology	Mittelbach ch 1
	Wed	19.aug	08.15-10.00	Lecture 2	BJG	Patterns in community composition in space and time	Ch 18 B&T
	Fri	21.aug	08.15-10.00	Seminar	SE/BJG	Intro to field courses	
	Fri	21.aug	12.15-14.00	Lecture 3	THR	Measuring biodiversity: Abundance and diversity of species	CH 18 B&T
35	Mon	24.aug	08.15-10.00	Lecture 4	SE	Food webs	Handed out literature
	Tue	25.aug	08.15-10.00	Lecture 5	SE	Abiotic factors in aquatic environments	Handed out literature
Field course Sletvik							
	Wed	26/8	09.00-12.00	Bus to Sletvik from Realfagbygget Meet in front of main entrance to Realfagbygget at 8.45			
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	Wed	02.sep	14.15-16.00	Lecture 8	PMP	Multivariate descriptions of communities	
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Schedule

- Lectures are not mandatory – but:
 - Material presented during the lectures forms basis for understanding the exercises and project work and also forms the exam curriculum
- Field course, labwork and project presentations are mandatory
 - Participation needs to be approved for attending exam
- Project report – 40% of grade
 - Contribution to project work and presentations by all students in group is mandatory
 - English
- Written exam – 60% of grade
 - English, Norwegian



The Project

- Training for later bigger projects in education or career – hands-on training with field, lab and data work
- We gather one common data set during the field course and follow-up with lab work
 - All students will contribute to all data collection and sample processing
- Each group of 4-6 students will focus on their own theme
 - Define why this is interesting - find and read related literature – develop hypotheses or research questions to test
 - analyse the data with assistance from supervisors/teachers
 - write up a report (abstract, introduction, methods, results, discussion, references)
 - present their project (powerpoint)



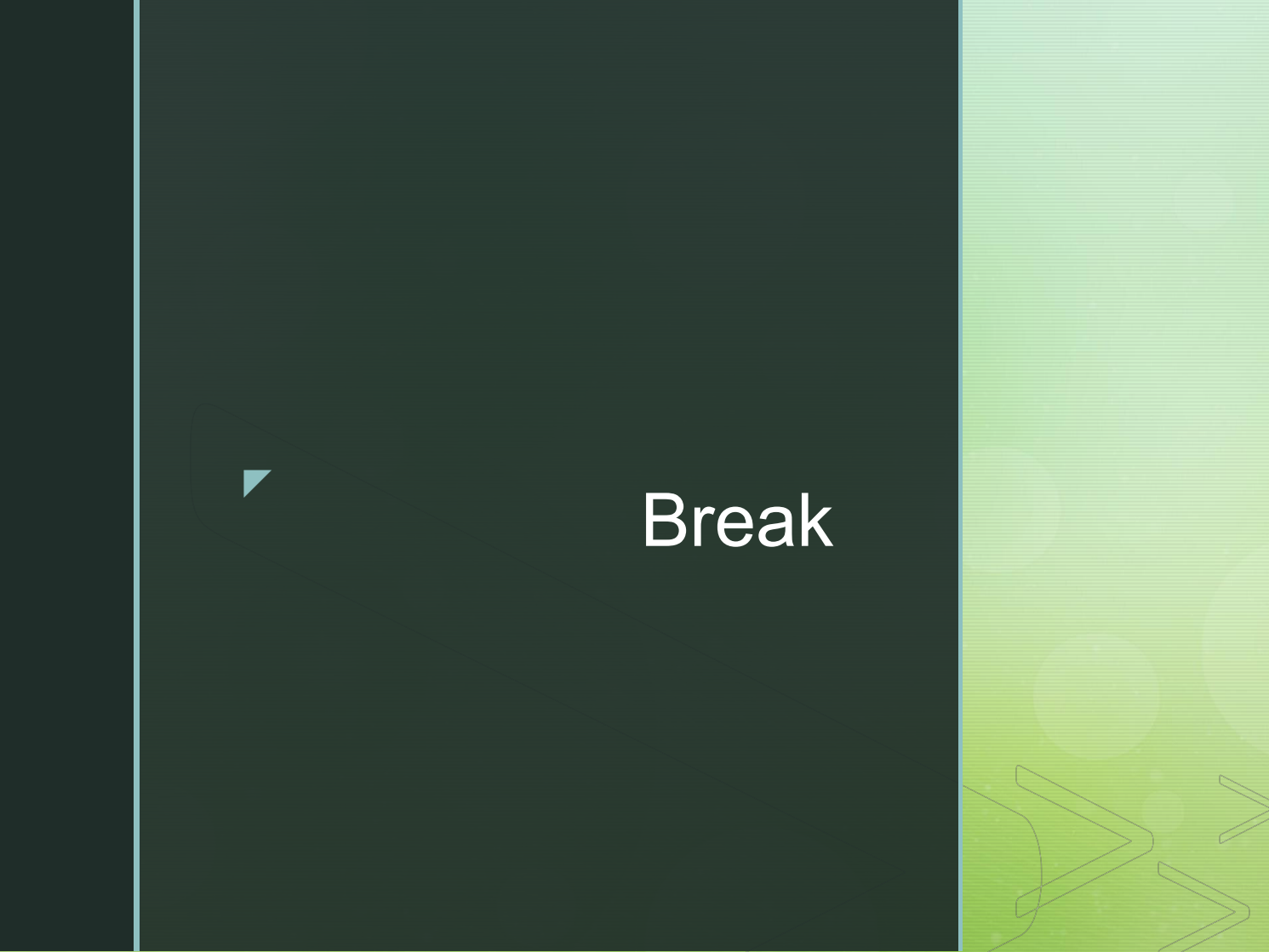
The purpose of the field course

- For you to experience the natural variation and explore with your own data the drivers that are shaping the nature – hence you need to understand a bit about Norwegian nature and how to investigate it



Student contribution to costs

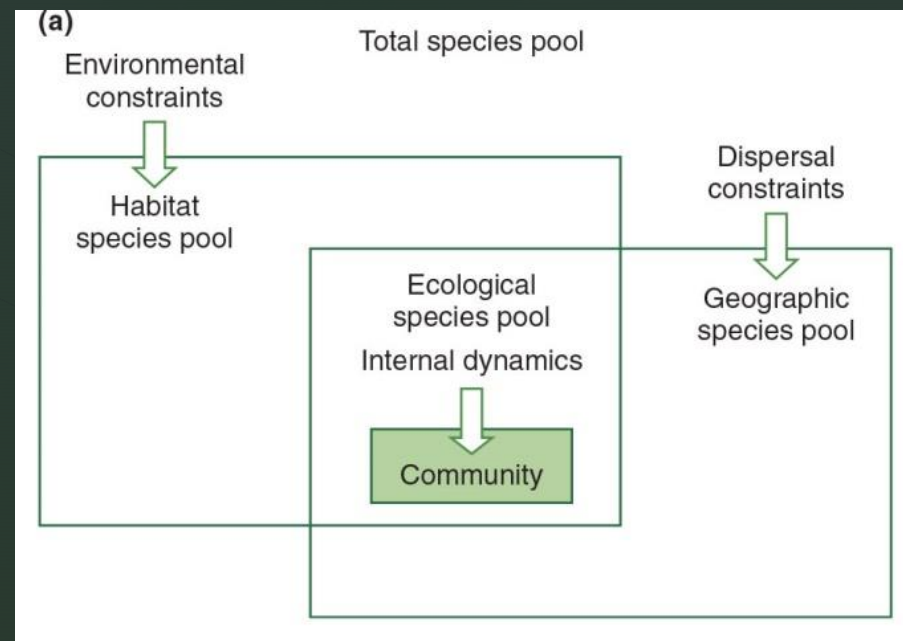
- Department of Biology requires a financial contribution towards the cost of food
- 660 NOK – Lena will put info on how to pay on Canvas
- We need you to fill in and hand us Field cards - Electronically please!
- But more on the logistics of the field courses on Friday



Break

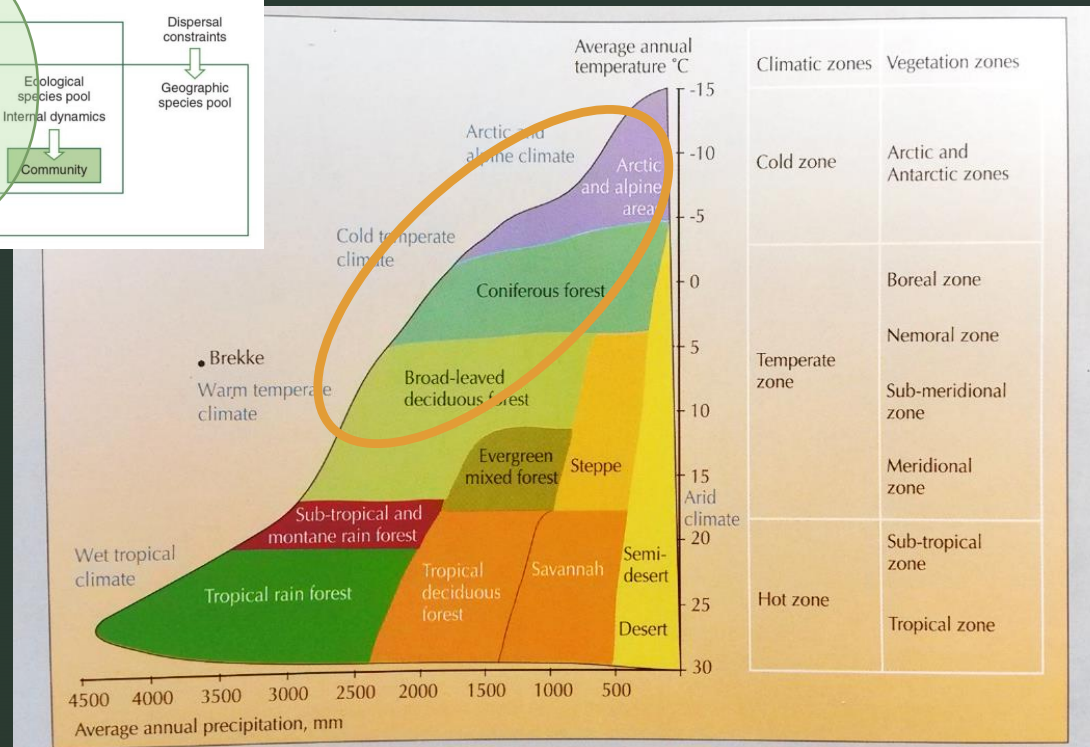
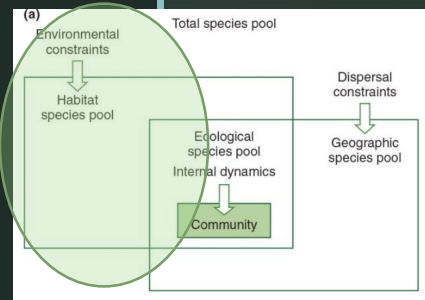


Communities - Coexistence in space





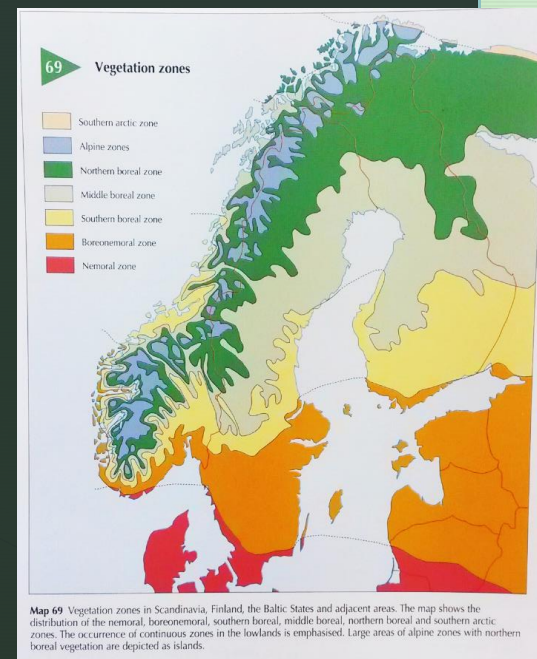
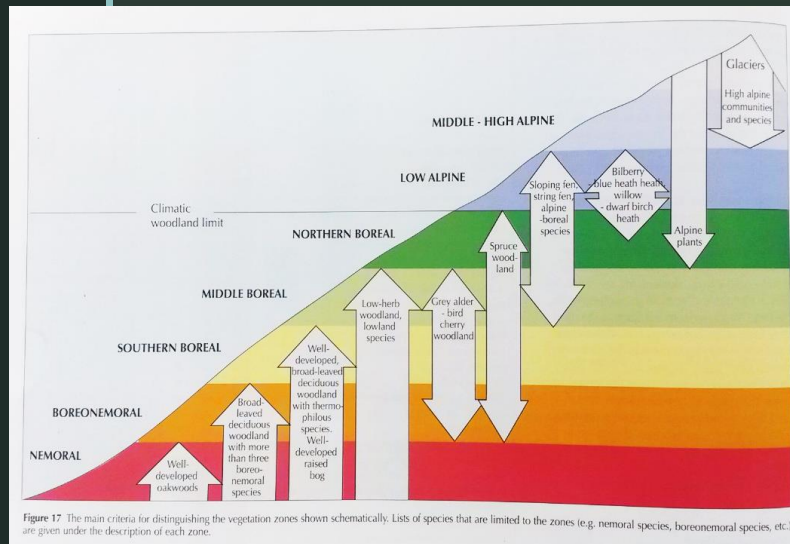
Environmental constraints: temperature and precipitation



Moen 1999 Vegetation: National Atlas of Norway

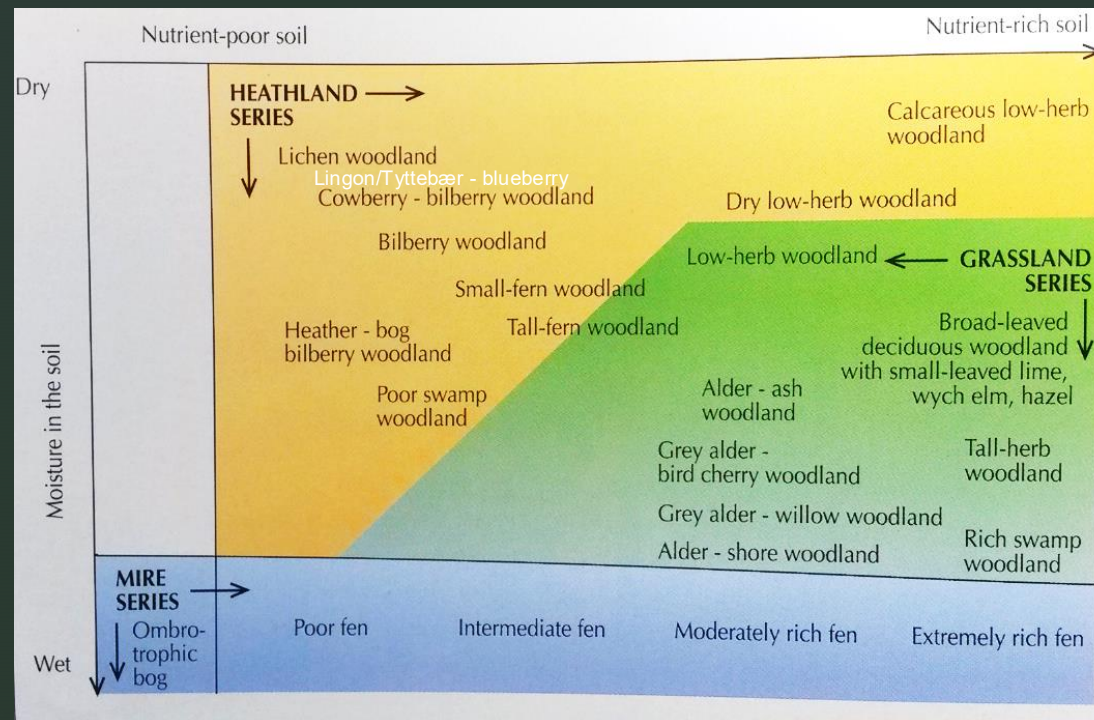


Elevation and latitude



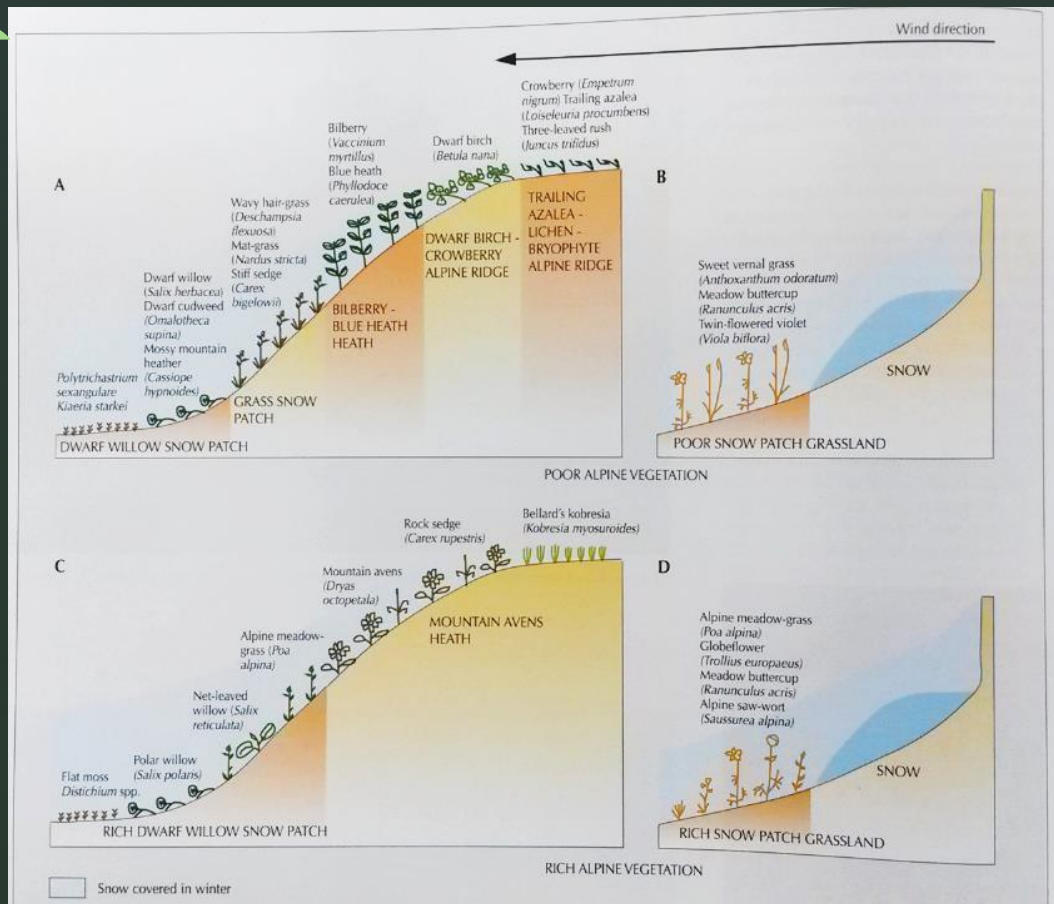


Environmental constraints below treeline: Nutrient status and soil moisture



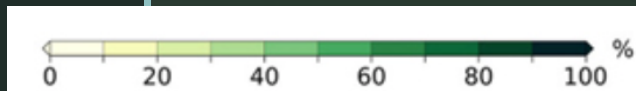
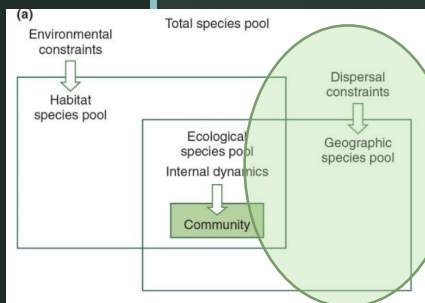


Smaller spatial-scales

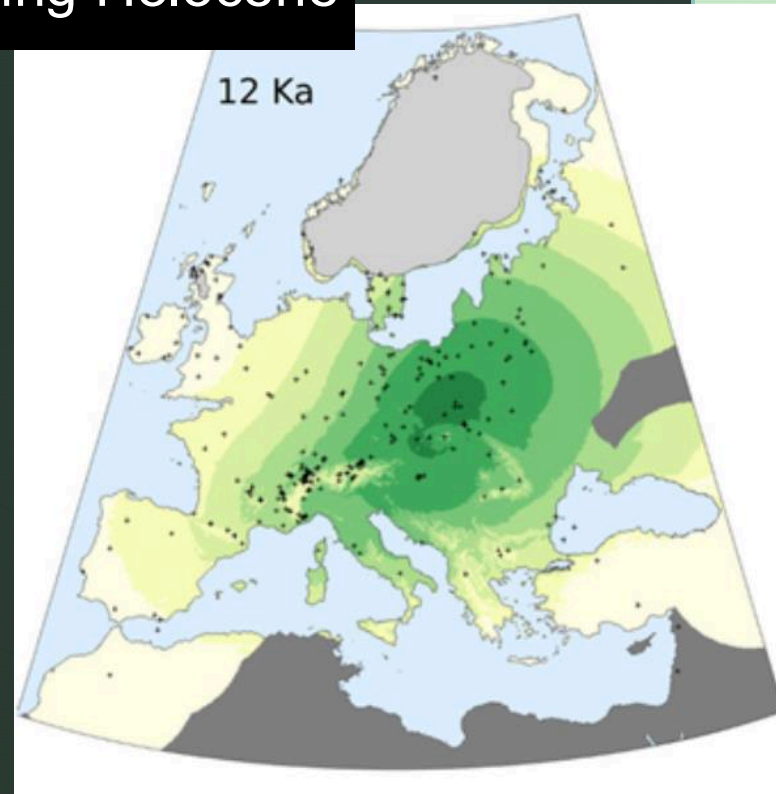




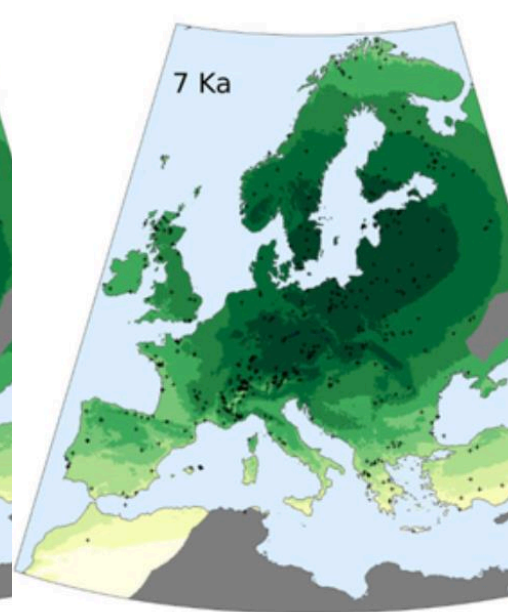
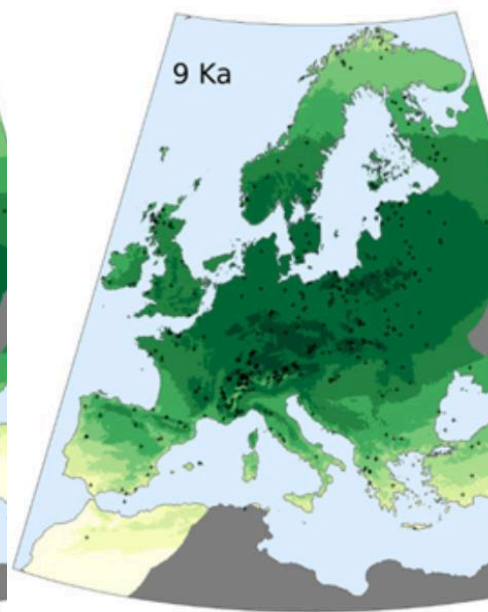
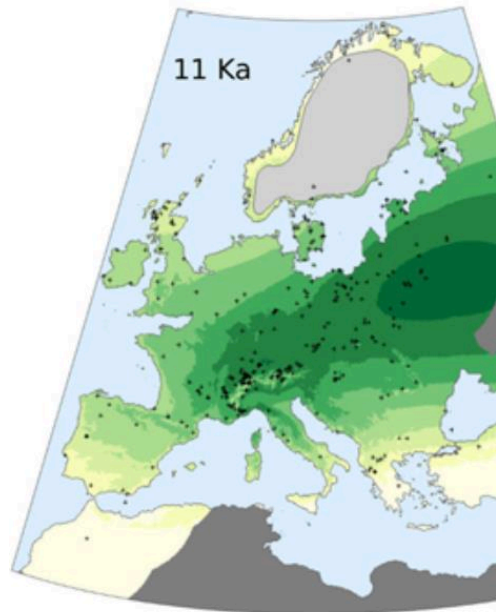
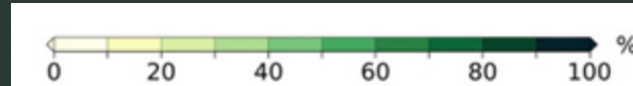
Ice retraction and forest expansion during Holocene

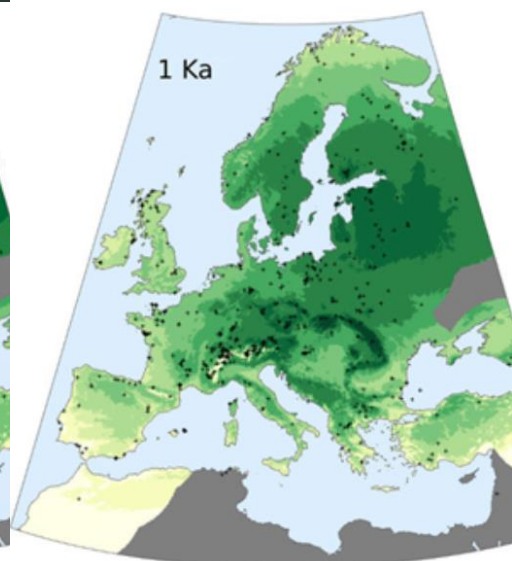
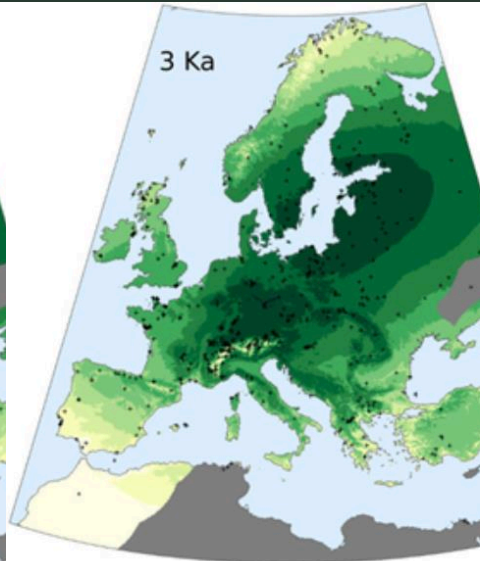
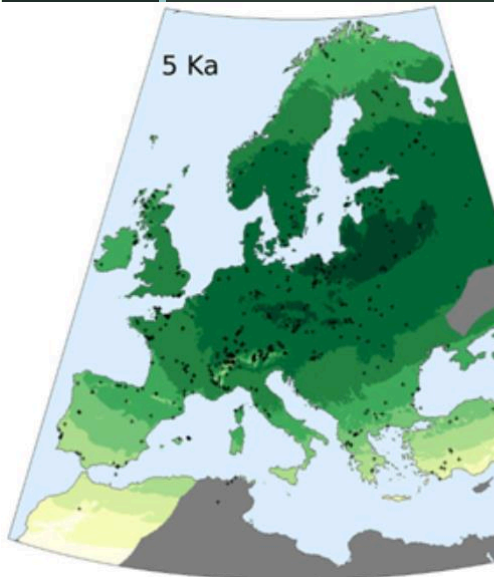
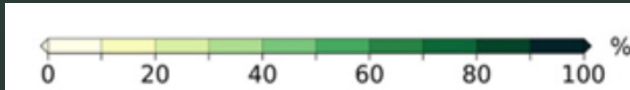


Dark grey = too little data
 Ka = 1000 years ago
 Dark grey means no data



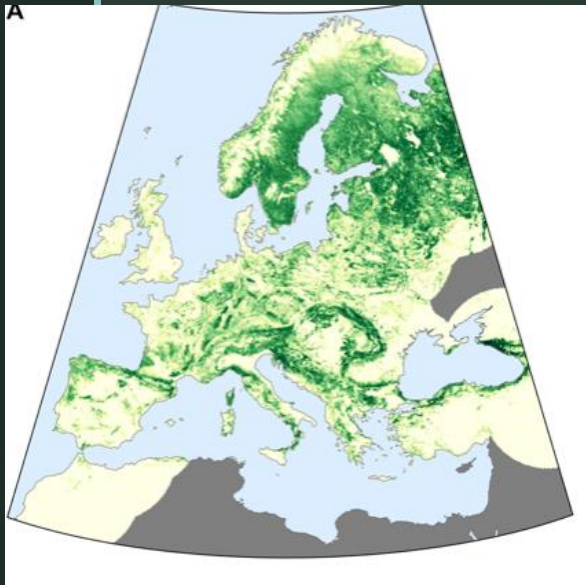
Front. Plant Sci., 08 March 2018
<https://doi.org/10.3389/fpls.2018.00253>



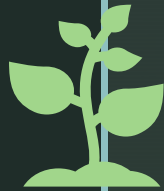




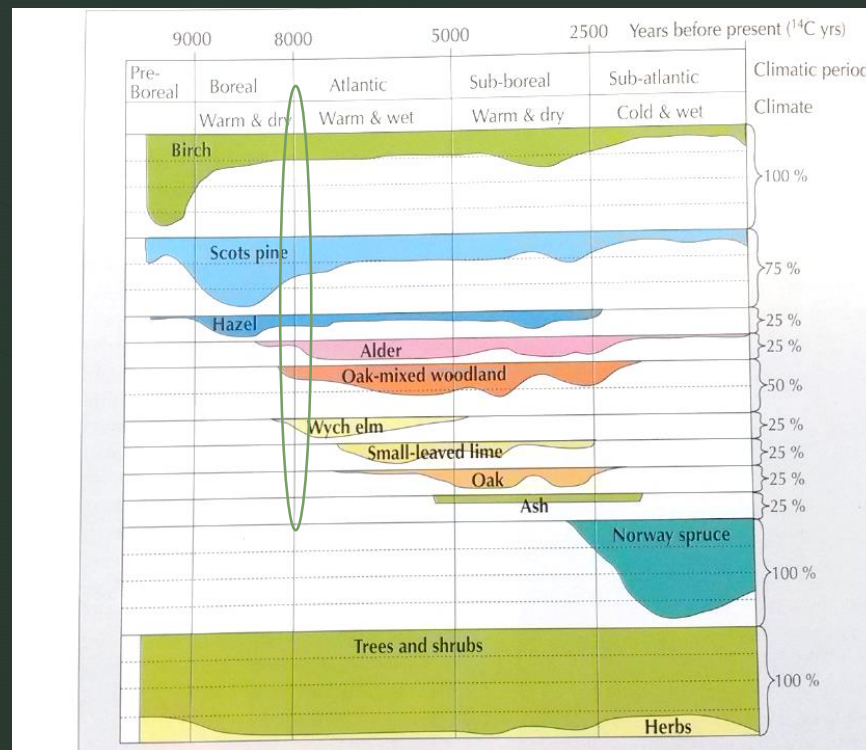
Drivers of the forest expansion?



- Climate
- Ice retraction
- Sea level dynamics – land upheaval
- Human settlements
- Of course the resulting vegetation is linked to other species in the landscape – but complexity in how?



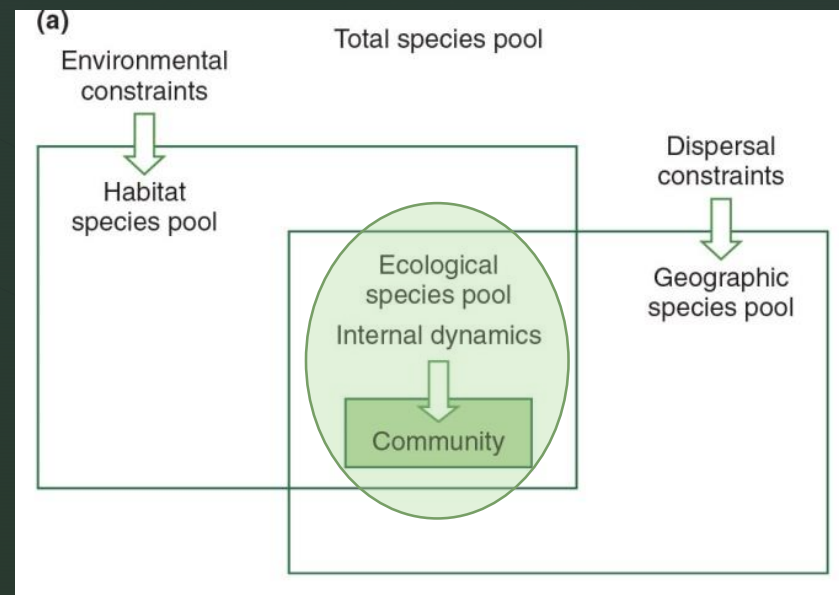
Communities - Coexistence in time



Moen 1999 Vegetation: National Atlas of Norway



Environmental and dispersal constraints are the foundation for the species assemblage – but then there are the internal dynamics – what is that??



We will study communities along an environmental gradient to understand drivers of community composition



Literature: Slides based on Mittelbach 2012 Ch1 and some
from Marchetti et al. 2023

Community and Ecosystem Ecology



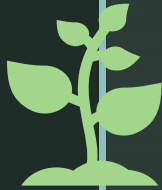
What is community ecology?

- An **ecological community** is the group of species coexisting in space and time
- Species may be interacting or not
- Communities are bound through sharing environment and potential interactions
- Community Ecologists study
 - **Patterns** in ecological communities – e.g. patterns of species richness, species composition, functional type dominance
 - **Interactions between species** – e.g. competition, facilitation, predation, parasitism
 - **Stability and dynamics** of communities
 - How communities are **affected by the abiotic environment** – e.g. nutrient availability, fire, climate



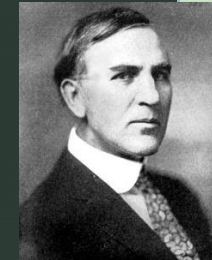
What is ecosystems ecology?

- The **combination of abiotic and biotic components**
- How do organisms **depend on abiotic variables**?
- How does the **biota affect the abiotic environment**?
- Important processes
 - Productivity
 - Nutrient cycling
 - Decomposition
 - Soil build up and stability
 - Carbon storage
 - Etc
- Often studied at larger spatial scales

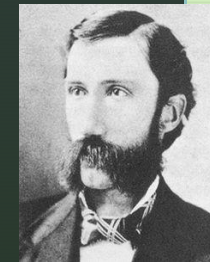


Brief history of community ecology Pre 1920s

- Frederic Clements (1916) suggests that plant associations are coherent units – super organisms – Plant communities follow a pattern of succession to a stable ‘climax community’
- Stephen Forbes (1887) suggests that a lake is a microcosm within which all organisms in a lake tend to function in harmony to create a system in balance
- Communities are seen as fixed entities
 - For a long time, the classification of communities into types was the aim of community ecology



Frederic Edward
Clements (1874-1945)

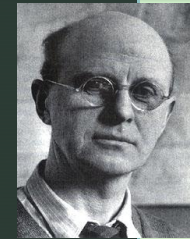


Stephen Alfred Forbes
(1844-1930)

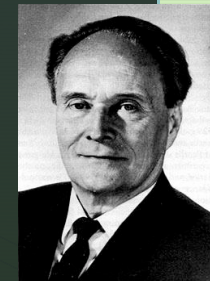


From (animal) populations to communities from 20s-50s

- Charles Elton opposed Forbes' ideas of balance in nature and published (1927) a number of important ideas including food webs, community diversity and community invasibility. '
- In the 1920s the mathematical ecologists Lotka and Volterra also developed models for competition and predation showing that ecological interactions could be modelled
- Gause (1934) tested these models and found they could predict ecological outcomes in bottle experiments (with protozoans, fruit flies, beetles, annual plants)
 - The competitive exclusion principle "Two species competing for the same resource cannot coexist with constant populations [if other factors remain



Charles Sutherland
Elton
(1900-1991)



Georgii
Frantsevich
Gause (1910-
1986)



The plant community ecology 1920s to 1950s

- Gleason counterargued the Clementsian view: Species have distinct ecological characteristics responding to environmental gradients.
- Gleason's ideas were initially ignored but his ideas led to the rejections of Clements theories (Whittaker 1956).
- However, Clements' theories regarding communities being distinct entities are still apparent in much of current ecology - Does a community exist as a unit, or simply a collection of species happening to be in the same place at the same time?
- Tansley introduced the ecosystem concept – highlighting the importance of energy and material flows between organisms and environment



Henry Allan
Gleason (1882-
1975)

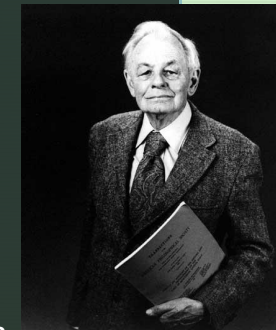


Sir Arthur George
Tansley
(1871-1955)

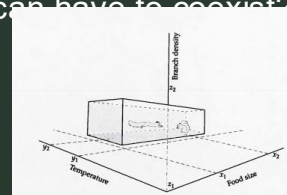
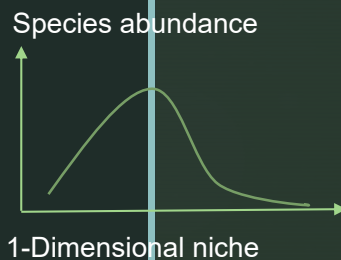


The ecological niche

- Hutchinson (1957) formalized the niche concept leading to the modern community ecology.
- A niche was quantified by including abiotic and biotic environments in an n-dimensional hypervolume.
- He defined the fundamental and the realised niche
 - The fundamental niche is (usually) wider than the realized niche, the latter limited by interactions such as competitive exclusion and predation).
 - His theory didn't include mutualism or commensalism.
- He also questioned (1959) how much niche overlap species can have to coexist?



George Evelyn
Hutchinson
(1903-1991)



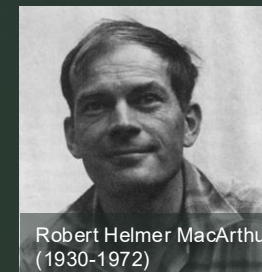
3-Dimensional niche

n-
Dimensional
niche

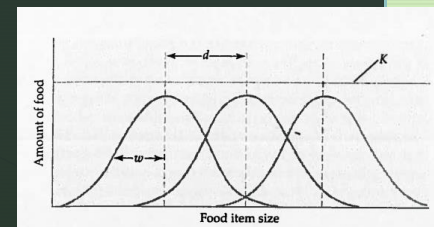


Beyond interspecific competition

- MacArthur's (1969) modelling showed that coexistence was possible where the ratio of the distance between resource optimum (d) and the standard deviation of resource utilization was >1
- However, such models often failed to describe the observed patterns in nature
 - Results were dependent on model assumptions and structure
 - Null models often explained equally well
- Competition and predation are not the only drivers of species diversity and distributions
 - Far more complexity in ecological communities



Robert Helmer MacArthur
(1930-1972)





Generalities in Community Ecology

- Lawton 1999 argued that there are very few universal laws in community ecology
 - Big pictures but messy details...
 - How do we even split one community from another?
- Stephen Hubbel suggested in 2000s that there are a lot of neutral dynamics out there. Systems where random patterns are more likely than the theoretical systems we suggest.
- 'Zombie ideas'
 - Ideas that refuse to die
 - 'The intermediate disturbance hypothesis should be abandoned' (Fox 2013)
- Don't despair – there are still a lot of patterns to study and try to tease out – and the research is highly needed 😊



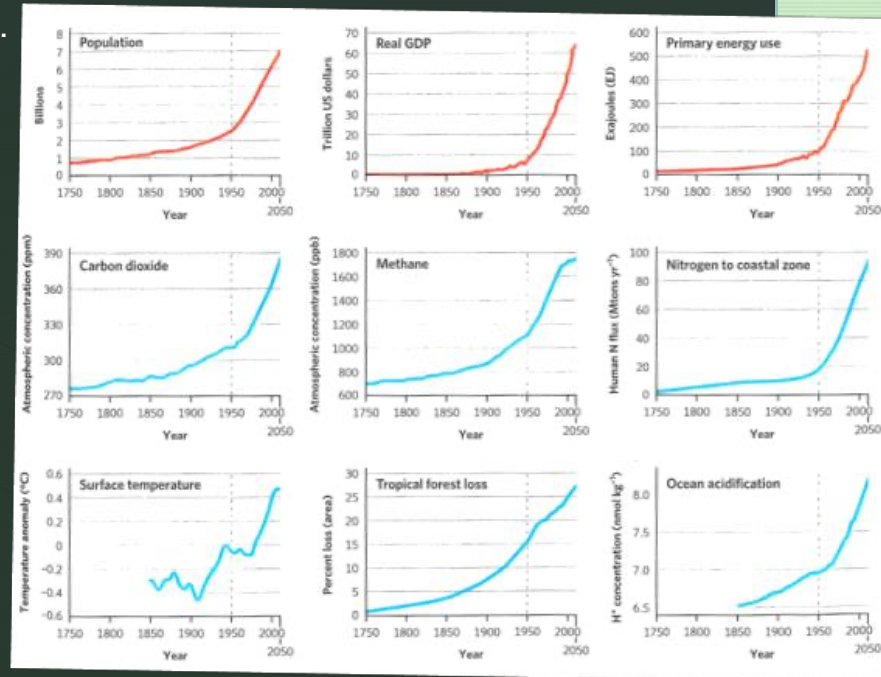
Professor Sir John Lawton





Some concepts from the book Marchants et al. 2003.
You can certainly observe real patterns and processes – but what are the drivers causing them?

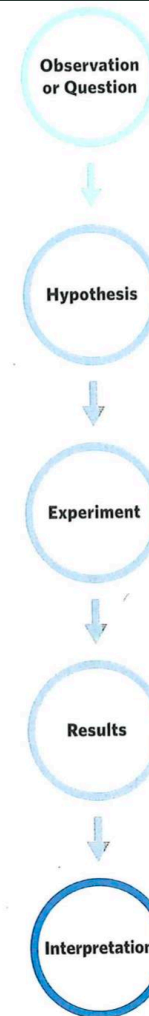
- You hypothesize drivers. Get ideas of how the world functions.
- These are ideas = concepts
- But they need to be tested for providing evidence that they are real.
- What is driving loss in biodiversity? Or forest loss? Climate change?





Experimental ecology and null hypothesis testing

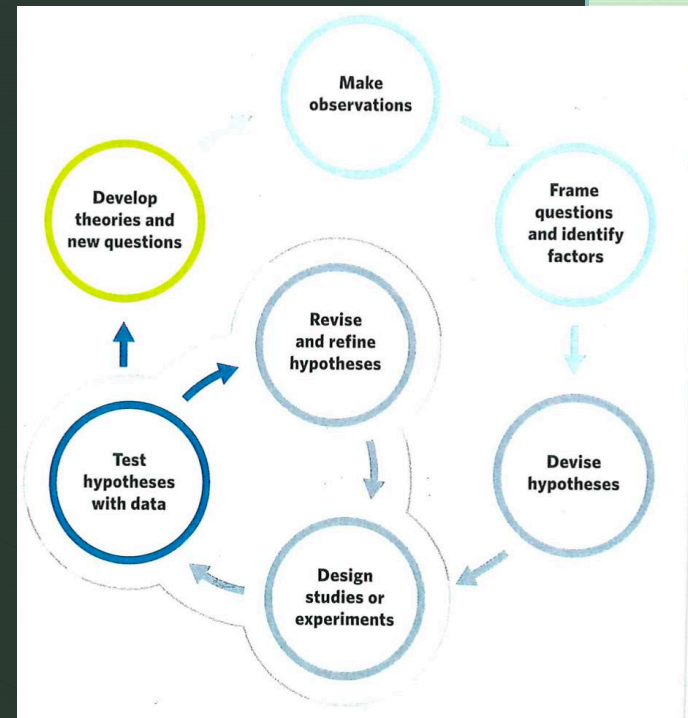
- Believe your observations but test the correlations and certainly any likelihood of causation
 - For evidence of any causation you need experiments
 - Test the hypothesis with experiments
 - If the result is not very accurate or wrong = no evidence
 - Improve the hypotheses or methodology
- You can also test it in terms of ground truthing or more observational studies
 - You develop a hypothesis – or a theory and then you go and survey/observe if this is so in nature
 - Or - You develop a theory based on one case and then you go and test if it is the same for other cases - always the same outcome = contribute to evidence
 - If not matching – reject your hypothesis – and make a new





The merry-go-round of ecologists

- The better observations/foundations the better opportunities for making the right assumptions for your theories and hypotheses





Summary of lecture

- Community ecology has developed from a Clementsian view where communities are seen as fixed entities developing towards or having reached climax stages towards a less rigid view where stochastic events are dominant, and succession is less predictable and developing along gradients.
- Communities vary in space and time
- Communities depend both on the abiotic environmental variation and the dispersal events on short and long time scales
- We do (community ecological and ecosystem) research by describing patterns we see and preferably testing if the patterns we see in nature can find some explanations that are likely causal (not only correlational). It may take a number of studies to confirm.
- Community ecology is very complex